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MANAGEMENT REVIEW

Management without theatre.



KPIS

TAKT TIME AND CYCLE TIME: THE BEAT OF DEMAND

Available time over demand, where takt came from, a line balanced from 64% to 93%, takt, cycle time and pitch, takt on building sites, how to time a process, and a worksheet.

5.5 months instead of the 11 planned for the exterior of a hospital in Sacramento, once the trades worked to a four-day takt.

FRANDSON ET AL., 2013

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Every operation runs on two clocks. The customer sets one: how often a unit has to be finished to keep up with demand. The process sets the other: how long each step really takes. This issue is about the two, and about what a manager does when they do not match.

The Lean Enterprise Institute defines takt time as the available time divided by customer demand. The idea came from German aircraft plants in the 1930s and spread through Toyota in the 1950s. On one line, eight unevenly loaded operators became two lines of three, and the balance rose from 64% to 93%. On a hospital in Sacramento, a four-day takt brought 11 planned months of exterior work down to 5.5. And measuring cycle time takes a stopwatch, a team that knows why, and at least ten timings per step.

Stiven Janaqi EDITOR · STIVEN CATALYST

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HOW TO READ THIS ISSUE

Figure
Every figure has its source and year at the foot of its page.

Our reading
Where the editors interpret rather than the research, it says so.

Practice
The steps and the card are proposals to try, not research results.

COVER STORY

The beat of demand

For the Lean Enterprise Institute, takt time is the available production time divided by customer demand. A plant that works 480 minutes a day for 240 units a day has a takt of two minutes. Takt, German for a precise interval such as a musical meter, is the heartbeat of a lean system.

1500s

Records of the arsenal of Venice describe a steady, takted output of merchant ships and warships.

1913

In Detroit, Ford brings the production line to the mass production of cars.

1930s

In the German aircraft industry, airframes move to the next station at a fixed interval: the takt.

1950s

Takt is widely used at Toyota; by the late 1960s, across its suppliers.

The interval is set by demand, not by the machines. According to the institute, Toyota typically reviews the takt of a process every month, with a smaller review every ten days.

OUR READING

Takt is not a speed you demand of people. It is the rate at which customers take what you make; every step is measured against it.

Sources: Lean Enterprise Institute, 2014; Shervin Haghsheno, Marco Binninger, Janosch Dlouhy & Simon Sterlike, Proceedings of the 24th Annual Conference of the International Group for Lean Construction (IGLC), 2016

Definition, aircraft and Toyota follow the LEI lexicon; Venice and Ford follow a review by Haghsheno et al. (2016). Accounts of how takt reached Japan through Mitsubishi rest on secondary sources, so we leave that step out.

THE NUMBERS

Eight people, two lines

In 2016 Curt Knoll, a former continuous-improvement manager, described a line from a recent project in IndustryWeek. Eight operators worked to a takt of 5.2 minutes; between the least and the most loaded there were exactly four minutes of work. The line balance was 64%.

LINE BALANCE, ONE PROJECT, 2016



16 → <2 min
operators' total waiting time, as the author adds it up: 87% less

+38%
output after the change

With training and process kaizen, the team ran two lines at a takt of 10.5 minutes each, with every operator under it. Work in process between operations fell from several units to one.

OUR READING

Waiting on a line rarely shows up in a report. It shows up in the balance chart, as the gap between the tallest bar and the shortest.

Source: Curt Knoll, IndustryWeek, 2016

One unnamed project, reported by the person who led it; the article does not say how the balance rate or the output were calculated. The figures are his.

THE MODEL

Takt, cycle and pitch

Takt comes from demand, cycle time from the process. LEI defines cycle time as the time it takes to make a part or complete a process, as timed by actual measurement. A process keeps pace when each step sits at or just under the takt.

01

Takt time

available time ÷ demand; set by the customer, the same for every step

02

Cycle time

measured at each step: an operator's full round of work, or a machine's work on one piece

03

Pitch

takt × pack-out quantity: a one-minute takt and boxes of 20 give a pitch of 20 minutes

HYPOTHETICAL EXAMPLE, PARCEL PACKING IN ONE SHIFT

Takt	450 minutes available, 900 parcels: 30 seconds
Cycle times	pick 26 s, check 11 s, pack 34 s, label 9 s
The gap	packing runs 4 s over takt: 900 parcels need an hour more than the shift has
People	80 s of work ÷ 30 s takt = 2.7: at least three, if the work is shared evenly

The bottleneck is visible before anyone is asked to work faster. The numbers are invented.

OUR READING

On a day when demand runs above forecast, the takt gets shorter. The steps that sat just under it are the first to fall behind.

Sources: Lean Enterprise Institute, Lean Lexicon: Takt Time, 2014; Lean Enterprise Institute, Lean Lexicon: Cycle Time, 2014; Lean Enterprise Institute, Lean Lexicon: Pitch, 2014

Definitions follow the LEI lexicon; the example and the count of people are the editors' arithmetic.

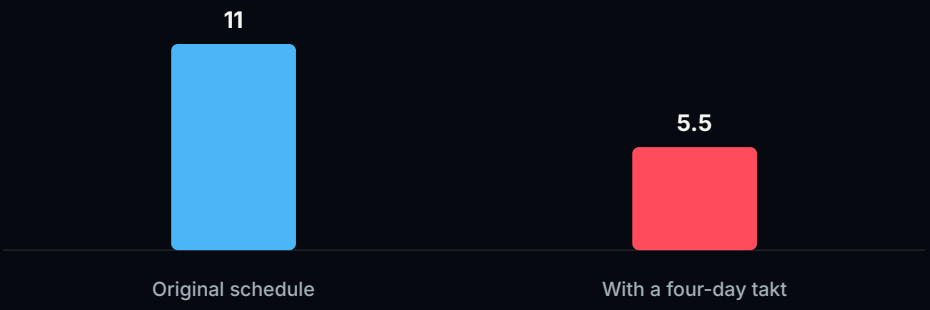
More in the essay — High-volume days: the standard under pressure

OUTSIDE THE FACTORY

Takt on a building site

Builders use takt too; a review by Haghsheno and colleagues names the Empire State Building (1930) as its first recorded use in building construction. In 2013 Adam Frandson, Klas Berghede and Iris Tommelein described the exterior of an eight-storey hospital in Sacramento, divided into zones that each trade had four days to finish.

EXTERIOR WORK ON A HOSPITAL IN SACRAMENTO, MONTHS



Framing was far slower than the other trades, so it got a second crew. The first three production plans were missed; every one after that was met. In Helsinki, a one-day takt per apartment cut the interior phase of a 42-apartment block by two months, nearly 30%, after a chaotic start and a partial restart.

“If the building process is not well-understood, then all the Takt time adds is stress.”

Sources: Shervin Haghsheno, Marco Binninger, Janosch Dlouhy & Simon Sterlike, Proceedings of the 24th Annual Conference of the International Group for Lean Construction (IGLC), 2016; Adam Frandson, Klas Berghede & Iris D. Tommelein, Proceedings of the 21st Annual Conference of the International Group for Lean Construction (IGLC), 2013; Joonas Lehtovaara, Iina Mustonen, Petteri Peuronen, Olli Seppänen & Antti Peltokorpi, Proceedings of the 27th Annual Conference of the International Group for Lean Construction (IGLC), 2019

Two single cases, reported by their authors. The Sacramento paper gives 5.5 months in its abstract and 5 in its results; we chart 5.5. The quote is the Sacramento team's lesson: at first they added time to the takt.

HOW IT IS MEASURED

Time it before you balance

Cycle time is measured, not estimated: the institute times an operator's cycle by direct observation. In IndustryWeek, Curt Knoll sets out how he prepares a time study before building the balance chart.

1. **Takt first**
   Work it out before you pick up a stopwatch.
2. **Walk the line both ways**
   Write every step on a time observation form.
3. **Tell the team why**
   You are not timing them to go faster; they work at their normal pace.
4. **Agree start and stop**
   In Knoll's own check, observers started and stopped the clock at different points.
5. **At least ten timings**
   Per step; twenty give an average you can trust more.

The institute's aim for the balance chart: each operator's work very nearly equal to, but slightly less than, takt. For a machine, count what comes with every piece: a 20-second cycle, 30 seconds to load and unload, and a 30-second changeover every 30 pieces give an effective cycle of 51 seconds.

Sources: Lean Enterprise Institute, Lean Lexicon: Cycle Time, 2014; Lean Enterprise Institute, Lean Lexicon: Operator Balance Chart, 2014; Curt Knoll, IndustryWeek, 2016

The steps follow Knoll (2016); the definitions and the machine example are the LEI lexicon's.

TOOL OF THE ISSUE

The takt worksheet

One process, one shift. Work out the takt from demand, time each step, and see which steps sit above the line before you add people or overtime.

01 Process and period which process; one shift, day or week

02 Available time minutes in the period, minus breaks and planned stops

03 Demand units the customer takes in the same period

04 Takt available time ÷ demand, in seconds

05 Cycle time per step at least ten timings each; note the spread, not only the average

06 Over takt and one change steps above takt; total work ÷ takt = people needed; what we move or improve first

Sources: Lean Enterprise Institute, Lean Lexicon: Takt Time, 2014; Lean Enterprise Institute, Lean Lexicon: Operator Balance Chart, 2014; Curt Knoll, IndustryWeek, 2016

A practice proposed by the editors, on the LEI definitions and Knoll's time study. The tool charts the timings and shows their spread.

Open the tool: Sigma & Control Chart stivencatalyst.com/tools/sigma-control-chart/

SOURCES AND METHOD

Every figure has a source.

The figures in this issue come from the sources below. The year shows how recent each one is.

Lean Lexicon: Takt Time
Lean Enterprise Institute, 2014
https://www.lean.org/lexicon-terms/takt-time/

History and Theoretical Foundations of Takt Planning and Takt Control
Shervin Haghsheno, Marco Binninger, Janosch Dlouhy & Simon Sterlike, Proceedings of the 24th Annual Conference of the International Group for Lean Construction (IGLC), 2016
https://iglc.net/Papers/Details/1297

Operator Balance: Why Waiting Is the Worst of the Wastes
Curt Knoll, IndustryWeek, 2016
https://www.industryweek.com/operations/continuous-improvement/article/21987519/operator-balance-why-waiting-is-the-worst-of-the-wastes

Lean Lexicon: Cycle Time
Lean Enterprise Institute, 2014
https://www.lean.org/lexicon-terms/cycle-time/

Lean Lexicon: Pitch
Lean Enterprise Institute, 2014
https://www.lean.org/lexicon-terms/pitch/

Takt Time Planning for Construction of Exterior Cladding
Adam Frandson, Klas Berghede & Iris D. Tommelein, Proceedings of the 21st Annual Conference of the International Group for Lean Construction (IGLC), 2013
https://iglc.net/Papers/Details/902

Implementing Takt Planning and Takt Control Into Residential Construction
Joonas Lehtovaara, Iina Mustonen, Petteri Peuronen, Olli Seppänen & Antti Peltokorpi, Proceedings of the 27th Annual Conference of the International Group for Lean Construction (IGLC), 2019
https://doi.org/10.24928/2019/0118

Lean Lexicon: Operator Balance Chart
Lean Enterprise Institute, 2014
https://www.lean.org/lexicon-terms/operator-balance-chart/

EDITORIAL METHOD

Each figure was checked for its year, its publisher and what exactly it measures. Where the publisher's page could not be opened, the figure was checked against independent summaries and is marked “via”. The editors' interpretation is marked “Our reading”. Figures that could not be confirmed are not in the issue.



MANAGEMENT REVIEW

Management without theatre.

Every issue, one management question, checked against the best research.

ALL ISSUES

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